

Mingyun Kang

Researcher in Construction Management & Automation

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scholar.google.com/citations?user=ZIYW21AAAAAJ

Education

Seoul National University of Science and Technology, Bachelor in Architectural Engineering, School of Architecture – Seoul, Republic of Korea Mar 2017 – Feb 2024

- Double Major: Business Administration
- GPA : 3.8 / 4.5

Seoul National University of Science and Technology, Master in Architectural Engineering (Integrated Bachelor-Master Program) – Seoul, Republic of Korea Mar 2024 – Aug 2025

- GPA : 4.22 /4.5
- Thesis : Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System (Advisor : Prof. Taehoon Kim)

Research Experience

Research Assistant, CONICT Lab, Seoul National University of Science and Technology – Seoul, Republic of Korea Dec 2021 – Aug 2025

Conducted research on computer vision-based automated construction supervision, 3D reconstruction, and human-robot collaboration at construction sites.

- Published 3 SCIE journal papers and 2 SCOPUS conference papers as first/co-author
- Filed 5 domestic patent applications related to construction automation and vision-based inspection
- Participated in 3 national R&D projects funded by NRF and the Ministry of Land, Infrastructure and Transport (MOLIT)

Post-Master's Researcher, Korea Institute of Civil Engineering and Building Technology (KICT) – Goyang, Republic of Korea Nov 2025 – present

Post-Construction Evaluation & Management Center, Conducted research on Post-Evaluation of SOC in Republic of Korea

Publications (SCIE)

Computer Vision-Based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System Dec 2024

Applied Sciences, 15(1), 125.
Mingyun Kang, Sebeen Yoon, Taehoon Kim
[10.3390/app15010125](https://doi.org/10.3390/app15010125) (Applied Sciences)

Approach to Enhancing Panoramic Segmentation in Indoor Construction Sites Based on a Perspective Image Segmentation Foundation Model Apr 2025

Applied Sciences, 15(9), 4875.
Juho Han, Sebeen Yoon, Mingyun Kang, Taehoon Kim
[10.3390/app15094875](https://doi.org/10.3390/app15094875) (Applied Sciences)

Automated Vision-based Location Monitoring for Jack Support

Jan 2025

KSCE Journal of Civil Engineering, 100374.

Sebeen Yoon, Mingyun Kang, Minhyung Kim, Taehoon Kim

[10.1016/j.kscej.2025.100374](https://doi.org/10.1016/j.kscej.2025.100374) (KSCE Journal of Civil Engineering)

Publications (SCOPUS)

Instance Segmentation of Exterior Insulation Finishing System using Synthetic Datasets

June 2024

ISARC Vol. 41, pp. 1176-1181. IAARC Publications.

Mingyun Kang, Sebeen Yoon, Juho Han, Sanghyeon Na, Taehoon Kim

[10.22260/ISARC2024/0152](https://doi.org/10.22260/ISARC2024/0152) (ISARC 2024 — International Symposium on Automation and Robotics in Construction)

Comparative Study of Structure from Motion on Construction Site

July 2025

ISARC Vol. 42, pp. 1395-1400. IAARC Publications.

Mingyun Kang, Sangmin Lee, Sebeen Yoon, Taehoon Kim

(ISARC 2025 — International Symposium on Automation and Robotics in Construction)

Publications (International Conference)

Computer Vision-Based Tile Counting Model for Automated Progress Monitoring

June 2023

CCC 2023, Keszthely, Hungary.

Mingyun Kang, Wooseok Lee, Junsang Yoo, Sebeen Yoon, Taehoon Kim

(Creative Construction Conference (CCC 2023))

A Preliminary Study on Automatic Jack Support Installation Interval Measurement Model

June 2023

CCC 2023, Keszthely, Hungary.

Sebeen Yoon, Mingyun Kang, Taehoon Kim

(Creative Construction Conference (CCC 2023))

Approach for Improving Panoramic Image Segmentation Quality on Construction Sites using SAM

June 2024

CCC 2024, Prague, Czech Republic.

Juho Han, Sanghyeon Na, Mingyun Kang, Sebeen Yoon, Taehoon Kim

(Creative Construction Conference (CCC 2024))

Publications (Domestic)

Computer Vision-based Adhesion Quality Inspection Automation Model for Exterior Insulation and Finishing System

Mar 2023

JKIBC, 23(2), 165-173. (KCI)

Sebeen Yoon, Mingyun Kang, Hyeonseung Jang, Taehoon Kim

(Journal of the Korean Institute of Building Construction (JKIBC))

Preliminary Study on Semantic Segmentation-based Quality Inspection Model for Exterior Insulation

Oct 2022

AIK, 42(2), 704-705.

Sebeen Yoon, Mingyun Kang, Byeongmin Lee, Taehoon Kim

(Proceedings of the Architectural Institute of Korea)

A Basic Study on Image-based Numerical Measurement Model for Construction Sites

May 2023

KIBC Spring 2023, 23(1), 287-288. [Best Paper Award]

Sebeen Yoon, Mingyun Kang, Taehoon Kim

(Proceedings of KIBC Spring Conference)

Applicability Enhancement of Deep Learning-based Computer Vision Models for 360° Panoramic Images KICEM 2023, Goseong, 205-206. Mingyun Kang, Junsang Yoo, Wooseok Lee, Taehoon Kim (Proceedings of KICEM Annual Conference)	Nov 2023
A Basic Study on Rebar Spacing Measurement Method Based on Single Image Information KIBC Spring 2024, Gunsan, 243-244. Cheolhun Park, Jieun Lee, Jihyeon Lee, Yunjae Jeon, Mingyun Kang, Taehoon Kim (Proceedings of KIBC Spring Conference)	May 2024
Performance Comparison Study on Image-based 3D Space Generation AIK Autumn 2024, Gyeongju, 974-975. Mingyun Kang, Sebeen Yoon, Taehoon Kim (Proceedings of Architectural Institute of Korea Autumn Conference)	Oct 2024
A Basic Study on 3D Reconstruction Performance Improvement Using 360° Camera KIBC Autumn 2024, 203-204. Mingyun Kang, Juho Han, Sebeen Yoon, Taehoon Kim (Proceedings of KIBC Autumn Conference)	Nov 2024
Construction of High-Resolution 360° Image Dataset Using Image Super-Resolution Technology National Student Conference 2024, 177-178. Minhyung Lee, Mingyun Kang, Taehoon Kim (Proceedings of National University Student Academic Conference)	Nov 2024
A Basic Study on Image Analysis-based Crackdown on Illegal Rooftop Buildings KIBC Spring 2025, Gyeongju, 309-310. Yeon Choi, Seonwoo Lee, Yujin Oh, Gangmin Bae, Mingyun Kang, Taehoon Kim (Proceedings of KIBC Spring Conference)	May 2025
Patents	
Automatic Supervision Method and Device for Adhesive Application Quality Domestic Patent Application No. 10-2023-0053756. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Sebeen Yoon, Mingyun Kang	Apr 2023
Automated Construction Site Supervision System Domestic Patent Application No. 10-2023-0125464. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Sebeen Yoon, Mingyun Kang	Sept 2023
Tile Quantity Counting System Domestic Patent Application No. 10-2023-0135013. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Sebeen Yoon, Mingyun Kang, Wooseok Lee, Junsang Yoo	Oct 2023
Method for Measuring Rebar Spacing Based on Single Image Information Domestic Patent Application No. 10-2024-0127210. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Mingyun Kang, Cheolhun Park, Jihyeon Lee, Jieun Lee, Yunjae Jeon	Sept 2024

Method for Detecting Illegal Rooftop Buildings Based on Video Image Analysis

Aug 2025

Domestic Patent Application No. 10-2025-0117608. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.

Taehoon Kim, Mingyun Kang, Seonwoo Lee, Yeon Choi, Yujin Oh, Gangmin Bae

Projects

Digital Mapping-based Platform Research Laboratory for Healthy Construction Life-cycle

June 2022 – Feb 2025

National R&D project exploring digital mapping-based platforms for the full construction lifecycle. Focused on computer vision applications for automated construction supervision.

- Vision-based adhesive application quality inspection for EIFS
- Tile quantity counting via computer vision

Development of Digital-based Architectural Supervision and Construction Automation Robot Technology

Apr 2022 – Dec 2026

Large-scale national R&D project (KRW 24,490M) funded by MOLIT aimed at developing digital architectural supervision systems and construction automation robots.

- Instance segmentation of insulation finishing systems using synthetic datasets
- Panoramic image segmentation for indoor construction sites

Development of Human-Robot Collaboration Technology and Integrated Operation System for High-Altitude Work at Construction Sites

Apr 2025 – Dec 2027

Ongoing national R&D project (KRW 17,000M) funded by MOLIT developing XR-based human-robot collaboration platforms for multipurpose high-altitude construction work.

- XR-based human-robot collaboration interface
- Integrated robot operation system for construction sites

Awards

Best Paper Award (최우수학술상)

May 2023

Awarded at the Spring Conference 2023 for the paper: A Basic Study on Image-based Numerical Measurement Model for Construction Sites

Korean Institute of Building Construction (KIBC)

Best Paper Award (최우수학술상)

May 2024

Awarded at the Spring Conference 2024 for the paper: A Basic Study on Rebar Spacing Measurement Method Based on Single Image Information

Korean Institute of Building Construction (KIBC)

Excellence Paper Award (우수학술상)

May 2025

Awarded at the Spring Conference 2025 for the paper: A Basic Study on Image Analysis-based Crackdown on Illegal Buildings on Rooftops

Korean Institute of Building Construction (KIBC)

Outstanding Thesis Award (우수논문상)

Aug 2025

Awarded for the Master's thesis: Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System

Seoul National University of Science and Technology

Languages & Skills

- Korean (Native) · English (Advanced) · Python (Advanced)

Hobbies

- Golf · Running · Snowboarding

Military Service

Sergeant (병장), Rifleman, Capital Defense Command, 1st Security Battalion, Republic of Korea Army – Republic of Korea Dec 2018 – Sept 2021

Completed mandatory military service as an infantry rifleman. Honorably discharged at the rank of Sergeant.

Certificates

Architectural Engineer License (건축기사) June 2025

TOEIC Speaking — Advanced Low (Score 160) Nov 2023

References

Professor Taehoon Kim